

9. [Maximum mark: 9]

- (a) Prove that $\tan\left(\theta - \frac{\pi}{4}\right) = \frac{\sin 2\theta - 1}{\cos 2\theta}$, where $\theta \neq \frac{(2n+1)\pi}{4}$, $n \in \mathbb{Z}$. [6]
- (b) Hence, or otherwise, solve $\frac{\sin x - 1}{\cos x} = \sqrt{3}$ for $0 \leq x \leq 2\pi$. [3]

[illegible]